



SIS 3204Y(5) – NLP

INTRODUCTION TO NATURAL LANGUAGE PROCESSING



WEEK 1: Introduction to NLP



Understanding how computers understand, process and generate human language.



Learn.

Explore core NLP concepts.



Apply.

Build with real-world examples.



Innovate.

Shape the future with NLP.

LEARNING OUTCOMES



• *By the end of this module, you will be able to:* •

L01



Explain the fundamental concepts, evolution, techniques, and applications of Natural Language Processing (NLP), including Large Language Models (LLMs).

L02



Apply appropriate NLP techniques and tools to preprocess, analyse, and model natural language data.

L03



Design, implement, and evaluate NLP solutions for real-world applications using contemporary AI and language technologies.

L04



Critically evaluate the ethical, societal, and technical challenges of NLP systems and recommend responsible AI practices.



WHAT IS NATURAL LANGUAGE?

Natural language is the way **humans** naturally communicate.



Examples of Natural Languages



English



Français



Español



中文 (Chinese)



العربية (Arabic)



Kreol Morisien



Characteristics of Natural Language



Ambiguous

One sentence can have multiple meanings.



Contextual

Meaning depends on the situation and context.



Expressive

Humans can express ideas, emotions and creativity.



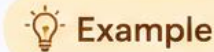
Dynamic

Language constantly evolves with time and culture.

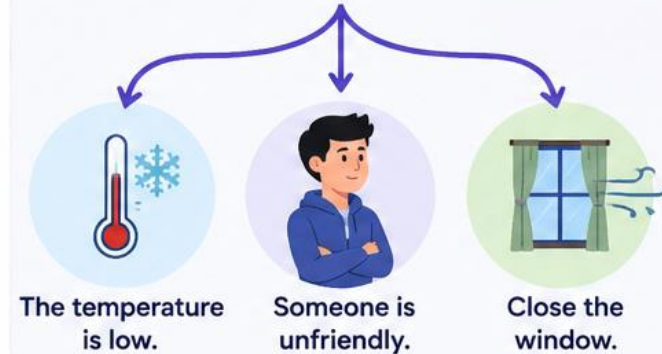


Why is it Challenging for Computers?

Computers follow **strict rules**.
Human language **does not**.



"It's cold."



Key Takeaway

Natural language is **beautiful, flexible** and powerful—but for computers, it is **complex and uncertain**.

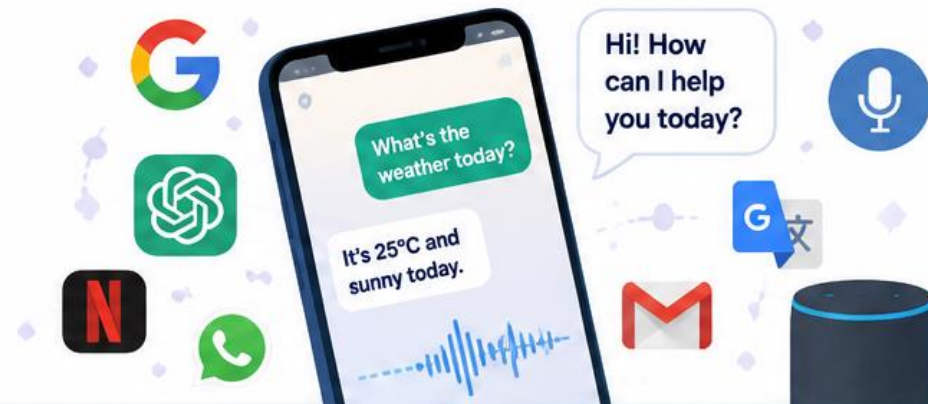


NLP IS EVERYWHERE

We use **NLP** every day without even realising it!



Natural Language Processing (NLP) powers many tools and services that make **our lives easier**.



APPLICATION		NLP TASK	APPLICATION		NLP TASK
	Search Engines e.g., Google Search	 Understanding queries, ranking results		Chatbots & Assistants e.g., ChatGPT, Kimi, Siri, Alexa	 Conversational AI, intent understanding
	Machine Translation e.g., Google Translate	 Translating text or speech		Voice Assistants e.g., Siri, Google Assistant	 Speech recognition, spoken language understanding
	Messaging Apps e.g., WhatsApp	 Smart reply, auto-correct, sentiment understanding		Writing Tools e.g., Grammarly	 Grammar checking, style improvement
	Email Filtering e.g., Gmail Spam	 Spam detection, priority inbox		Banking & Finance e.g., Chatbots, Fraud detection	 Text classification, intent detection
	Entertainment e.g., Netflix, YouTube	 Recommendations, video search, captions		Healthcare e.g., Clinical Notes Analysis	 Information extraction, text summarisation



KEY MESSAGE

From the apps we use to the services we rely on, NLP is the invisible technology behind **human-computer communication**.



NLP IN THE REAL WORLD

From ideas to impact — NLP is transforming every industry.



NLP helps machines **understand, analyse and generate** human language, enabling **smarter decisions** and **better experiences** across all sectors.

“

NLP is not just technology — it's solving **real problems** for real people!

”



1 HEALTHCARE



- Analyse clinical notes and reports
- Extract symptoms, conditions and drugs
- Support medical diagnosis & triage

Examples: Clinical documentation, medical chatbots, patient support, drug information extraction

2 FINANCE



- Detect fraud & suspicious transactions
- Analyse customer feedback
- Automate document processing

Examples: Fraud detection, risk assessment, customer service bots, KYC document processing

3 EDUCATION



- AI tutors & personalised learning
- Automatic essay scoring
- Q&A from textbooks and lectures

Examples: Smart tutoring, automated grading, plagiarism detection, content summarisation

4 BUSINESS



- Customer support & chatbots
- Analyse surveys & reviews
- Automate reports and documents

Examples: Virtual assistants, customer feedback analysis, document summarisation, meeting minutes

5 GOVERNMENT



- Improve citizen services with chatbots
- Analyse public feedback and social media
- Policy & document analysis

Examples: E-governance, policy analysis, grievance redressal, multilingual citizen communication

6 CYBERSECURITY



- Detect phishing emails & malicious content
- Analyse threat intelligence
- Monitor dark web text data

Examples: Spam filtering, threat detection, incident response, security log analysis

★ WHY IT MATTERS



Enhances human **productivity**



Improves **decision making**



Creates better **user experiences**



Drives innovation and **growth**



NLP is the **language bridge** between people and technology.



ONE TECHNOLOGY. ENDLESS POSSIBILITIES. REAL-WORLD IMPACT.

NLP powers smarter systems that **understand** us — and help us **do more**.



WHAT WILL YOU LEARN?

Your *NLP Journey* – From Foundations to Intelligent Language Systems



By the end of this module, you will have the skills and knowledge to build and understand **real-world NLP systems**.

Your goal:
Build intelligent systems that understand and generate human language!



SEMESTER 1 – NLP FOUNDATIONS & MODERN MODELS



1. NLP Foundations
Language, text, tokens, tokenisation, preprocessing



2. Text Representation
Bag-of-Words, TF-IDF, word embeddings



3. Neural Networks for NLP
Feedforward, RNNs, LSTMs, GRUs



4. Transformers
Attention mechanism, self-attention, encoder-decoder



5. Large Language Models
Architecture, pre-training, prompting, fine-tuning



6. Speech Processing
Speech to text (ASR), text to speech (TTS)



7. Conversational Agents
Chatbots, dialogue systems, intent recognition



SEMESTER 2 – ADVANCED NLP APPLICATIONS



1. Part-of-Speech Tagging
Tagging words with their grammatical roles



2. Language Models
N-gram models, perplexity, probability estimation



3. Sentiment Analysis
Detecting opinions and emotions in text



4. Information Retrieval
Search engines, indexing, ranking, retrieval models



5. Text Mining
Pattern discovery, clustering, association rules



6. Machine Translation
Statistical and neural machine translation



7. Text Summarisation
Extractive and abstractive summarisation



8. Responsible NLP
Ethics, fairness, bias, privacy, and safety



By the end of this module, you will be able to:



Understand human language deeply



Build models that process and generate text



Work with speech and conversations



Apply NLP to solve real-world problems

WHAT MAKES HUMAN LANGUAGE DIFFICULT?

Human language is rich, dynamic and full of nuances.



Humans understand language naturally.
Computers do not.

While humans effortlessly grasp meaning, intention and context, computers struggle with all of it.

“Behind every simple sentence, there is incredible complexity.”



Human language is:



CONTEXTUAL

Meaning depends on the situation, the speaker, and what came before or after.



AMBIGUOUS

The same words or sentences can have multiple meanings.



EMOTIONAL

Tone, feelings and intentions influence the true meaning of a message.



FIGURATIVE

We use idioms, metaphors, sarcasm and jokes to express ideas.



CULTURAL

Language is shaped by culture, traditions, shared experiences and background knowledge.



DYNAMIC

Language evolves over time. New words, slang and meanings emerge constantly.



KEY MESSAGE

The same sentence can have different meanings depending on context.
This is what makes NLP one of the most challenging problems in AI.



Our goal: Teach computers to understand language as **intelligently** as humans do.

AMBIGUITY IN LANGUAGE

— One sentence can have multiple meanings.



Words and sentence structure alone are often not enough.
We rely on context to understand the real meaning.

Humans use
context.
Computers
need algorithms.



EXAMPLE 1

"I saw the man with the telescope."

Who has the telescope?

A



The man
The telescope belongs to the man.

B



You
You have the telescope.

EXAMPLE 2

"The chicken is ready to eat."

Who is eating?

A



The chicken
The chicken is going to eat.

B



Someone
Someone is going to eat the chicken.

EXAMPLE 3

"Visiting relatives can be annoying."

Who is visiting?

A



We are visiting
We are the ones who are visiting our relatives.

B



They are visiting
Our relatives are visiting us.



KEY MESSAGE

Humans use context,
world knowledge and
experience to choose
the right meaning.



DISCUSS

Can you think of
other sentences
with multiple
meanings?



Takeaway

Without context, the same sentence can lead to different interpretations.
This is one of the biggest challenges in Natural Language Processing.



Context is everything!

CONTEXT CHANGES EVERYTHING

— The same words can have completely different meanings.

EXAMPLE 1

“It’s cold.”

Different situations, different meanings.

1



The temperature is low.

(Weather)

2



Please close the window.

(Request)

3



Someone is unfriendly or distant.

(Personality)

4



The air conditioning is too strong.

(Environment)

“Meaning is never in the words alone. It’s in the context.”



EXAMPLE 2

“Nice job!”



Genuine compliment

Said with a smile, positive tone, and supportive context.

VS



Sarcasm or irony

Said with a different tone, facial expression or in a negative situation.



DISCUSSION QUESTION

How do humans know which meaning is correct?



Context



Experience



World Knowledge



Intention & Tone



KEY MESSAGE

Context helps us choose the right meaning from many possibilities.



WHY IT MATTERS FOR NLP

Computers only see words (or speech). They must learn to **use context** to **pick the right meaning** — this is one of the core challenges in NLP.

REAL EXAMPLE USING CHATGPT

Can AI understand context?



The same word can have completely different meanings depending on the context around it.

Words don't have meaning. Words have meaning in context.

“

AI gets closer to the right meaning when we give it more context.

”



Let's try with the word “bank”

1 Ask ChatGPT:



The bank is full.

Likely answer (financial context):



A financial bank is full of customers.
(Refers to a place where people do banking.)

2 Ask ChatGPT:



The fisherman sat near the bank.

Meaning changes completely:



The fisherman sat near the river bank.
(Refers to the land beside a river.)

3 Ask ChatGPT:



The children played on the river bank.

Different again:



The children played on the land beside the river.
(Again, “bank” refers to the edge of the river.)



LEARNING POINT

The word “bank” has different meanings:



Financial institution



River bank (edge of water)



Context decides the meaning



TAKEAWAY

Without context, AI may guess incorrectly.
With context, AI gets much closer to the right meaning.

This is why context is at the heart of NLP.



HUMANS vs COMPUTERS

Understanding language is easy for humans, extremely hard for computers.



Humans bring a lifetime of knowledge and experience.
Computers only receive **text or speech**.

“
Computers see
words.
Humans see
meaning.”



HOW HUMANS UNDERSTAND LANGUAGE



Context – what is happening around us
(e.g., location, time, previous events)



Experience – our past helps us interpret
new situations



Emotion – feelings influence how we
interpret words



Culture & values – shared beliefs and
social norms



Tone & non-verbal cues – voice, facial
expressions, body language



Common sense – world knowledge
and reasoning

VS



Same sentence,
very different
resources!

WHAT COMPUTERS RECEIVE



Text (characters)
e.g., “It’s cold.”



Speech (audio signal)
e.g., sound waves



Images / Visual data
e.g., pixels



Limited background knowledge
from training data



Rules & algorithms
learned from data



Everything else
must be
inferred by
the machine!



KEY MESSAGE

Human understanding is rich and multi-dimensional.
NLP aims to help computers get closer to **human-like understanding**.



WHAT IS NATURAL LANGUAGE PROCESSING (NLP)?



Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to **understand, interpret, generate** and **interact** using **human language**, whether in text or speech.



TEXT



SPEECH



HUMAN

What is the weather today?



It is 25°C and sunny today.



AI SYSTEM



TEXT



SPEECH

NLP POWERS MANY REAL-WORLD APPLICATIONS



Chatbots & Virtual Assistants



Language Translation



Search Engines



Speech Recognition



Text Summarisation



Sentiment Analysis



Email Spam Filtering



Information Extraction



WHAT CAN NLP DO?



NLP enables computers to **understand**, **interpret**, **generate** and **interact** using human language.



1. UNDERSTAND TEXT

NLP helps computers understand the meaning of words, sentences and documents.



2. GENERATE TEXT

NLP models can generate human-like text, summaries, stories, answers and more.



3. TRANSLATE LANGUAGES

NLP allows computers to translate text or speech from one language to another.



4. ANSWER QUESTIONS

NLP systems can understand questions and provide accurate and relevant answers.



5. SUMMARISE TEXT

NLP can automatically create short and meaningful summaries of long documents.



6. RECOGNISE SPEECH

NLP converts spoken language into text (STT – Speech to Text).



7. GENERATE SPEECH

NLP converts text into natural-sounding speech (TTS – Text to Speech).



8. HOLD CONVERSATIONS

NLP enables chatbots and virtual assistants to have meaningful and contextual conversations.



9. RETRIEVE INFORMATION

NLP helps search engines and systems find the most relevant information from large collections.



10. ANALYSE OPINIONS

NLP can detect emotions, sentiment and opinions in text (positive, negative or neutral).



Key Takeaway

NLP empowers machines to process, understand and generate human language, opening the door to **countless real-world applications**.

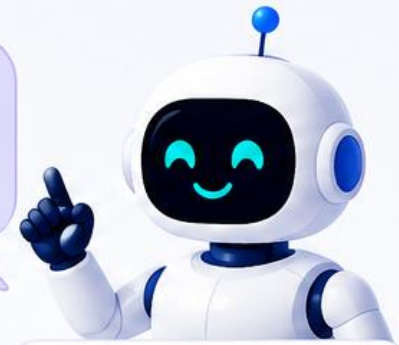


THINK ABOUT TODAY...

From the moment you woke up,
how many **NLP** systems did you use?



You interact with **Natural Language Processing (NLP)** more times in a day than you realize!



Google Search

Understands your query and finds the best answers.



ChatGPT / Kimi

You ask questions, get explanations, and generate ideas.



WhatsApp

Smart reply, voice messages, and translation.



Gmail

Filters spam, suggests replies, and organizes your emails.



Google Translate

Translates text or speech between different languages.



Voice Search

Speaks your words and understands what you mean.



Google Maps

Understands places, addresses and gives directions.



YouTube

Searches videos, understands captions and recommendations.



Netflix

Recommends shows based on titles, descriptions and reviews.



TikTok / Social Media

Understands captions, comments and suggests content you like.



KEY MESSAGE

Almost every digital service you use relies on **NLP** to understand, interpret and generate human language.



NLP TASKS vs NLP APPLICATIONS

— All powerful AI applications are built from a small set of core NLP tasks.



NLP breaks down human language into meaningful pieces, understands it, and generates useful outputs.
Applications combine multiple tasks to solve real-world problems.

“ Think of NLP Applications as what we see. NLP Tasks are what makes them work behind the scenes! ”



NLP TASKS (Building Blocks)



TEXT PROCESSING

- Tokenization
- Cleaning
- Normalization
- Part-of-Speech Tagging



SPEECH PROCESSING

- Speech Recognition (STT)
- Speech Synthesis (TTS)



UNDERSTANDING & ANALYSIS

- Classification
- Named Entity Recognition
- Sentiment Analysis
- Information Retrieval
- Question Answering



GENERATION OF TEXT

- Text Generation
- Summarisation
- Translation
- Dialogue Generation



NLP APPLICATIONS (What We See)



ChatGPT



Siri / Alexa



Gmail
(Spam Filter)



Google
Translate



Netflix
(Recommendations)



Google Search
(Understanding Queries)



Chatbots
& Assistants



Grammarly
(Writing Assistant)



KEY IDEA

No single task can do everything.

Real-world systems combine multiple NLP tasks working together.

This is what makes NLP so powerful and useful!



Big Picture:

NLP Tasks are the building blocks.
NLP Applications are the solutions that improve our daily lives.



Build



Combine



Create Impact

MAJOR NLP TASKS

The core building blocks behind every NLP application



All NLP systems perform these fundamental tasks — individually or together.

“
Different tasks.
Different goals.
Same foundation:
human language!
”



 NLP TASK	 WHAT IT DOES	 EXAMPLE	 REAL-WORLD USE CASE
 Text Classification	Assigns a label or category to text.	“This email is spam.”	 Email spam filtering
 Sentiment Analysis	Detects emotions or opinions in text.	“This product is amazing!” (Positive)	 Product reviews, social media monitoring
 Machine Translation	Converts text from one language to another.	“Hello” → “Bonjour”	 Google Translate, multilingual communication
 Speech Recognition	Converts spoken language into text.	(User says) “Open YouTube” (Text output) Open YouTube	 Siri, Alexa, voice assistants
 Text Generation	Creates human-like text based on input.	“Once upon a time...”	 ChatGPT, content creation, automatic replies
 Text Summarisation	Condenses long text into shorter form.	Long article → Short summary	 News summarisation, research assistance
 Question Answering	Finds the answer to a question from text.	Q: Who is the Prime Minister of Mauritius? A: Pravind Jugnauth.	 Chatbots, search engines, virtual assistants
 Named Entity Recognition (NER)	Identifies names of people, places, organisations, etc.	“Satya Nadella is CEO of Microsoft.” (Person, Organisation)	 Information extraction, data organisation

REMEMBER

 These tasks are the foundation of all NLP solutions.

 Real applications use multiple tasks working together.

 Master the tasks, build amazing solutions!



ONE LANGUAGE. **MANY TASKS.** ENDLESS POSSIBILITIES.
This is the **power** of NLP!



Understand



Analyse



Generate



Solve Problems

A DAY WITH NLP

— NLP is everywhere—working quietly to make your day easier!



From the moment you wake up to the time you sleep, you interact with **multiple NLP systems** every single day.

“

You may not see NLP, but NLP sees, hears, understands and helps you—every day!

”



How NLP is part of your everyday life...



KEY TAKEAWAY

NLP helps you:

- ✓ Speak to machines
- ✓ Find information
- ✓ Filter & organise content
- ✓ Get smart suggestions
- ✓ Understand other languages
- ✓ Make better decisions



You use NLP many times a day without even thinking about it!



THINK ABOUT IT!

How many NLP systems have you used today?

More than you think!



MEET TODAY'S AI SYSTEMS

—  **Powerful tools. Different strengths. Choose the right one for the right task!**



Large Language Models (LLMs) are advanced AI systems trained on massive data. Each system has **unique strengths**, features and **best use cases**.

“

No single AI is the best for everything. The smartest users know **WHICH** tool to use and **WHEN!**

”



ChatGPT

by OpenAI

MAIN STRENGTH



General-purpose assistant

BEST FOR

- Explaining concepts
- Writing & editing
- Brainstorming ideas
- Coding help

EXAMPLE USE

Draft an email, explain a topic, or write Python code.



Claude

by Anthropic

MAIN STRENGTH



Writing quality & reasoning

BEST FOR

- Long-form writing
- Complex reasoning
- Analysis & insights
- Careful responses

EXAMPLE USE

Write reports, analyse text, or reason through complex problems.



Gemini

by Google

MAIN STRENGTH



Google ecosystem & multimodal

BEST FOR

- Real-time information
- Images & documents
- Google Workspace
- Research & learning

EXAMPLE USE

Search the web, analyse images, or summarise YouTube videos.



Kimi

by Moonshot AI

MAIN STRENGTH



Long documents & research

BEST FOR

- Very long documents
- Summarisation
- Reading & research
- Academic support

EXAMPLE USE

Summarise papers, books, or long reports efficiently.



Copilot

by Microsoft

MAIN STRENGTH



Productivity & programming

BEST FOR

- Coding assistance
- Microsoft Office
- Data analysis
- Automation tasks

EXAMPLE USE

Write code, create Excel formulas, or design PowerPoint slides.



Perplexity

by Perplexity AI

MAIN STRENGTH



AI-powered search

BEST FOR

- Finding accurate info
- Cited answers
- Research queries
- Fact checking

EXAMPLE USE

Get answers with sources, references and links.



Different tools. Different strengths.

The best results come from using the **right tool** for the **right task**!

How to choose?

- ✓ Understand your goal
- ✓ Experiment and compare
- ✓ Know each tool's strength
- ✓ Combine tools smartly



DISCUSSION TIME

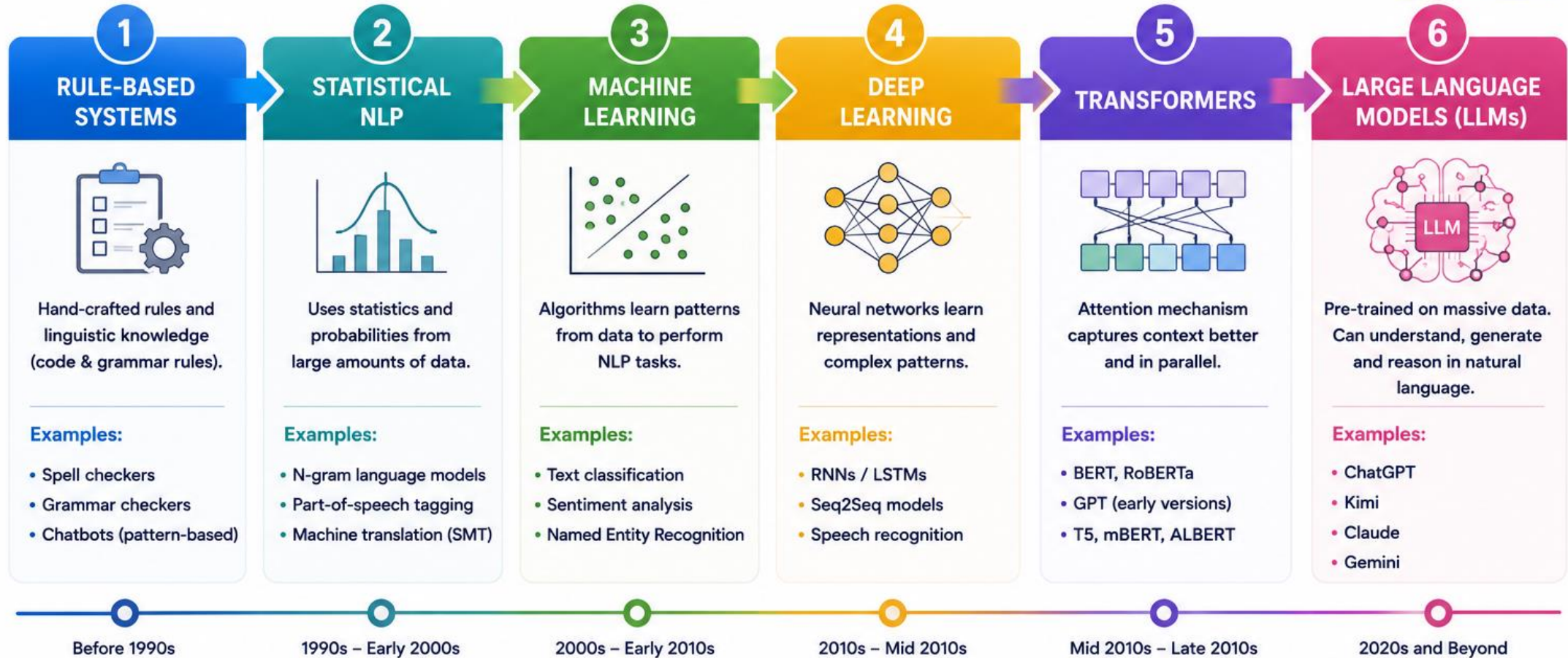
Which of these AI systems have you used before?
What do you like most about it? What could be better?



THE EVOLUTION OF NLP

From rules to reasoning – NLP has come a long way!

“
Better language
technology,
smarter impact.”



KEY TAKEAWAY

NLP has evolved from manual rules to data-driven models, to deep neural networks, to powerful LLMs that can **understand, generate and interact** like never before.



Human
Language

+



AI
Technology

=



Real-World
Impact

RULE-BASED NLP

The early days: When rules and dictionaries powered NLP



In the early years (1950s–1980s), NLP systems relied on **hand-crafted rules** and **linguistic knowledge** coded by experts. The goal was to teach computers how language works using **explicit instructions**.



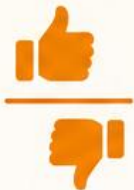
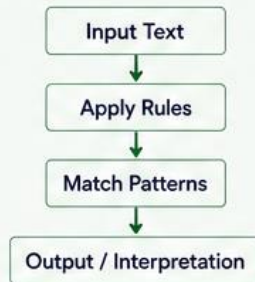
HOW IT WORKED

- Build dictionaries of words and their properties.
- Write grammar rules to analyze sentence structure.
- Use pattern matching and logic to understand text.
- Combine rules, heuristics and expert knowledge.



KEY COMPONENTS

- Dictionaries & Lexicons
- Grammar Rules
- Morphological Rules
- Heuristics & Pattern Matching
- Expert Systems



ADVANTAGES & LIMITATIONS

- ✓ **Advantages:** Easy to understand, explainable, works well in narrow and well-defined domains.
- ✗ **Limitations:** Difficult to scale, brittle to new data, cannot handle ambiguity and real-world variation well.



EXAMPLE: PART-OF-SPEECH TAGGING

A rule-based system might use simple rules like:

```
IF word ends with "ing" THEN Verb (VBG)
IF word ends with "ed" THEN Verb (VBD)
IF word is in {the, a, an} THEN Determiner (DT)
IF word is in {and, but, or} THEN Conjunction (CC)
IF word is capitalized AND not first word
  THEN Proper Noun (NNP)
```

Example Sentence:



This approach worked for simple cases, but struggled with complex, real-world language.



Rule-based NLP laid the foundation, but the need for data-driven methods led to the next era.

Next: Slide 34 – Machine Learning in NLP



MACHINE LEARNING IN NLP

Learning from data instead of writing rules



Instead of manually creating rules, we let computers **learn patterns from data**.
These models **improve automatically** as they see more examples.



THE IDEA

Give the computer many examples and the correct answers.



The computer learns the patterns and makes predictions on new data.



COMMON NLP TASKS SOLVED USING MACHINE LEARNING

- Spam Detection
- Sentiment Analysis
- Language Identification
- Named Entity Recognition
- Machine Translation
- Text Classification

These tasks were very hard with rules, but ML made them possible!

HOW IT WORKS (SIMPLIFIED)



1. DATA

Collect a large set of texts with known answers (called “labels”).



2. TRAINING

The model learns patterns and relationships from the data.



3. PREDICTION

The model uses what it learned to predict the answer for new, unseen text.



4. EVALUATION

We check how well it performs and improve the model if needed.

EXAMPLE: SPAM DETECTION



Training Data:

Thousands of emails labeled as “Spam” or “Not Spam”.



The model learns patterns like:

- certain words
- word frequency
- sentence structure
- sender behavior



New Email:

“Win cash now! Click here!!!”

Model Prediction: SPAM



COMMON MACHINE LEARNING ALGORITHMS USED IN NLP



Naïve Bayes

Simple and fast; great for text classification (e.g., spam detection).



Decision Trees

Easy to understand; works well with structured features.



Support Vector Machines (SVM)

Effective for high-dimensional text data.



Logistic Regression

Predicts probabilities; widely used for classification tasks.



Random Forest

Combines many trees; robust and handles complex patterns.



KEY TAKEAWAY

Machine Learning allowed NLP to move from handcrafted rules to **data-driven systems**, making models more **accurate**, **scalable** and **adaptable** to real-world language.



ML marked a turning point—teaching computers to learn from data opened the door to modern NLP.

Next: Slide 35 – Deep Learning Revolution →

THE DEEP LEARNING REVOLUTION IN NLP

From features to representations: Models that learn automatically



Deep learning allows models to learn **complex patterns** and representations **automatically** from large amounts of data, eliminating the need for manual feature engineering.

KEY BREAKTHROUGHS



NEURAL NETWORKS

Inspired by the human brain. Learn patterns using layers of interconnected neurons.



WORD EMBEDDINGS (Word2Vec)

Represent words as dense vectors in a continuous space. Capture semantic similarity.



SEQUENCES & CONTEXT

Models that understand word order and context in sentences.

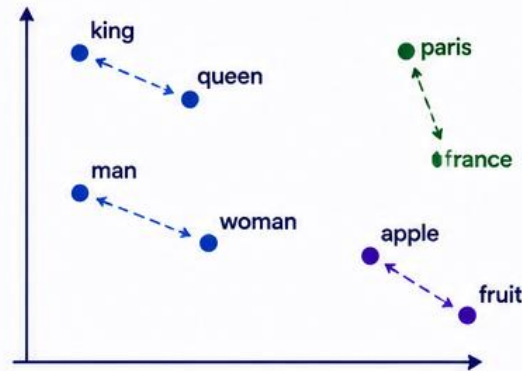


BETTER PERFORMANCE

Deep learning models achieve state-of-the-art results on many NLP tasks.

EXAMPLE: WORD2VEC

Words with similar meaning are closer in vector space.

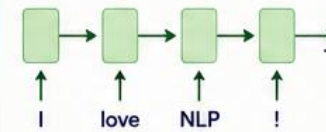


$\text{king} - \text{man} + \text{woman} \approx \text{queen}$
paris is closer to france, apple to fruit ...

MODELING SEQUENCES

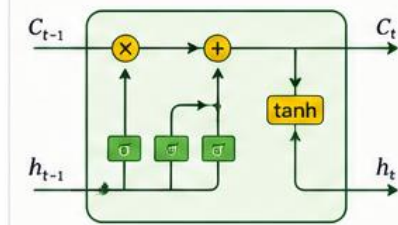
RNN (Recurrent Neural Network)

Processes sequence one word at a time, keeping context in memory.



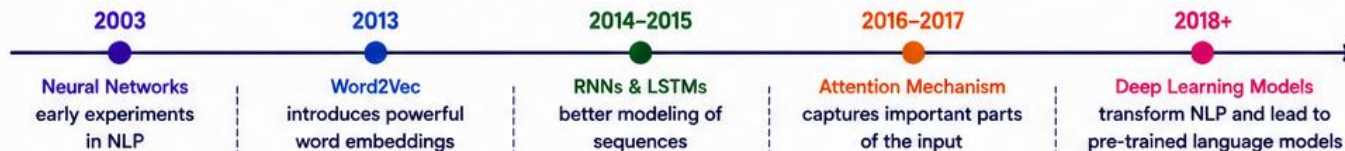
LSTM (Long Short-Term Memory)

Better at remembering long-term dependencies and context.



Impact: Deep learning models learn rich representations and context, leading to major improvements in NLP applications.

MILESTONES IN DEEP LEARNING FOR NLP



REAL-WORLD IMPACT

- More accurate machine translation
- Better chatbots and virtual assistants
- Improved sentiment and spam detection
- Powering today's large language models (LLMs)



KEY TAKEAWAY

Deep learning shifted NLP from manual rules to **automatic representation learning**, enabling models to understand language **more like humans**.



Deep learning unlocked the power of data, context and representation—paving the way for Transformers and LLMs.

Next: Slide 36 – Transformers Changed Everything →

TRADITIONAL NLP vs MODERN NLP

A shift from rule-based systems to powerful, data-driven models



NLP has evolved from handcrafted rules and limited capabilities to **intelligent, flexible, and general-purpose** models.

ASPECT	TRADITIONAL NLP	MODERN NLP
 Approach	Rule-based and handcrafted by experts. 	Data-driven and learned from examples. 
 Data Requirement	Works with small, task-specific datasets. 	Needs large-scale datasets to learn effectively. 
 Task Handling	Task-specific systems. One system = one task. 	General-purpose models. One model = many tasks. 
 Context Understanding	Limited context. Mostly keyword or short patterns. 	Understands long-range context and complex relationships. 
 Performance	Limited accuracy. Struggles with ambiguity and variation. 	High accuracy. Better at handling ambiguity, noise and variation. 
 Scalability	Difficult to scale. Rules become complex and hard to maintain. 	Highly scalable. Models improve as data and compute grow. 
 Examples	Chatbots with fixed rules, spell checkers, grammar tools, keyword search. 	ChatGPT, Claude, Gemini, machine translation, sentiment analysis, summarization, Q&A. 



KEY TAKEAWAY

Modern NLP has transformed how machines understand and generate language—from rigid rule systems to **flexible, intelligent models** that learn and adapt.



Modern NLP is not just better—it's a fundamental shift in how we build intelligent language systems.

Next: Slide 38 – NLP Today: The Modern Landscape



HOW DOES NLP ACTUALLY WORK?

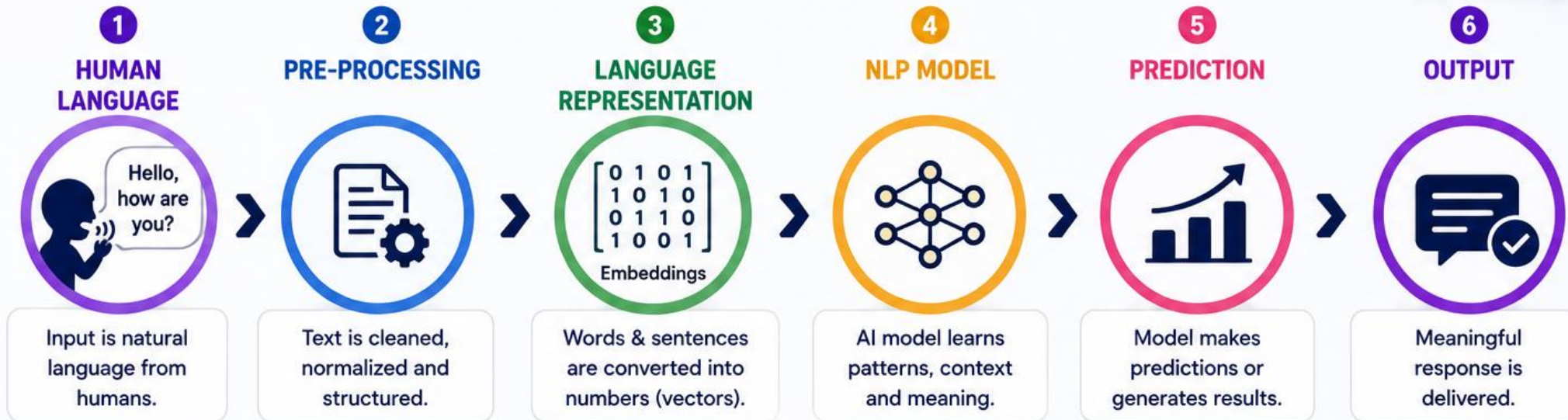
From human language to intelligent responses



Students often ask:
“How does ChatGPT understand me?”

Answer: Through a sequence of processing steps.
This is the **NLP Pipeline**.

“ NLP systems take human language, process it step by step, and produce meaningful outputs. ”



EXAMPLE

User:
“Translate: Hello”



BIG IDEA: NLP is not magic. It is a **sequence of steps** that turns human language into **intelligent understanding** and action.



We will explore **each step in depth** this semester!

THE NLP PIPELINE

Every NLP application follows a similar journey from input to output.

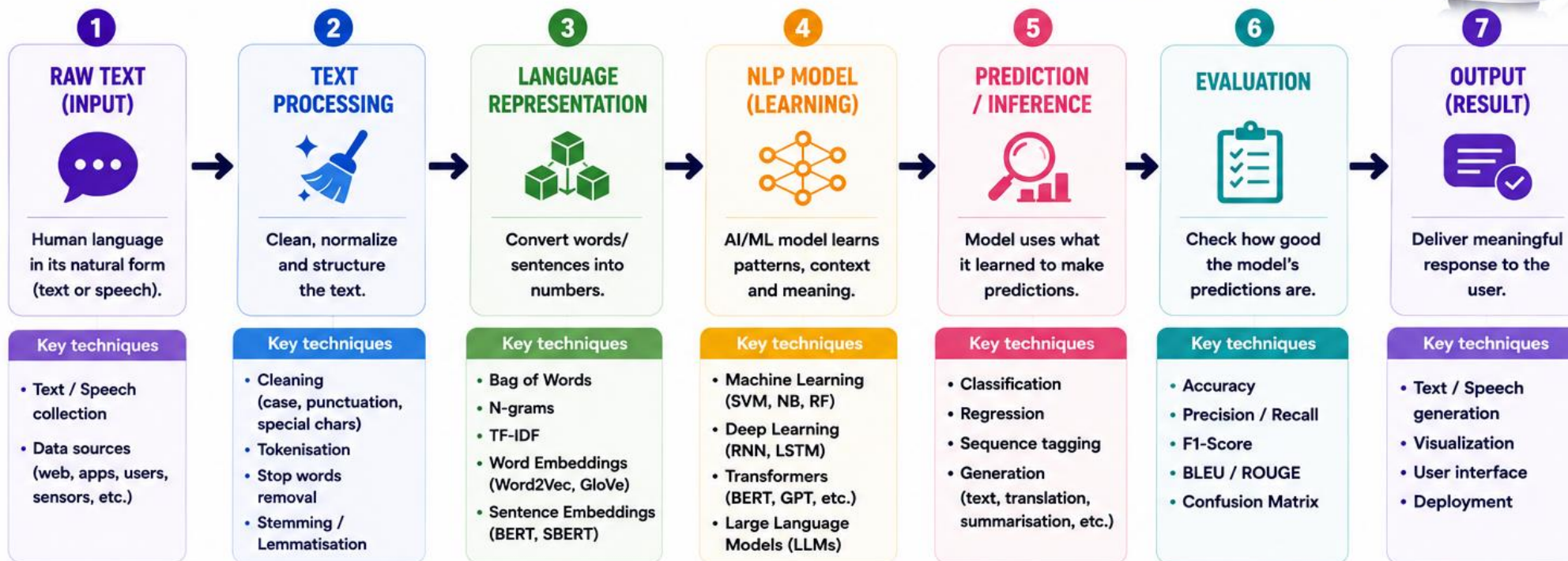


Different tasks (translation, sentiment, summarisation, QA, etc.) may use different models, but the **overall pipeline** is the same.

“

Think of NLP like a factory that takes raw human language and turns it into something **intelligent** and **useful**!

”



IMPORTANT TAKEAWAY
No matter what the task is, **every NLP system** follows this pipeline!

Examples of tasks using this pipeline:



Translation



Sentiment Analysis



Question Answering



Text Summarisation



Chatbots



Information Retrieval



Everything we study this semester **connects** to one or more stages of this pipeline.



Master the pipeline, master NLP!

STEP 1: TEXT PROCESSING

We clean and prepare raw text so that machines can understand it.



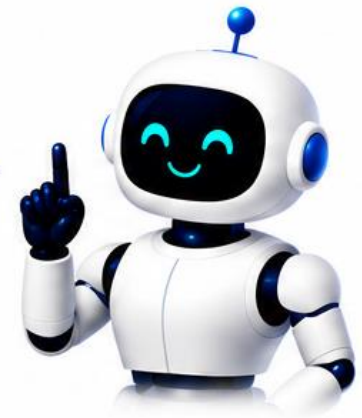
Raw text from the real world is messy and inconsistent.
Text processing **standardizes** and **structures** it.

“

Good data in,
good insights out!

Clean text leads to
better AI.

”



EXAMPLE

Raw Input:

I LOVE Natural Language Processing!! 😊 It's amazing, isn't it?



1. CLEANING

Remove extra spaces,
punctuation, emojis,
convert to lowercase

i love natural language
processing its amazing
isnt it



2. TOKENISATION

Split text into smaller
units (tokens)

['i', 'love', 'natural',
'language', 'processing',
'its', 'amazing', 'isnt', 'it']



3. STOP WORD REMOVAL (Optional)

Remove common words
that add little meaning

['love', 'natural',
'language', 'processing',
'amazing']



4. STEMMING / LEMMATISATION (Optional)

Reduce words to their
base form

['love', 'natural',
'languag', 'process',
'amaz']



5. NORMALISED TEXT

Final structured text
ready for the next step

love natural language
process amazing



WHY IS TEXT PROCESSING IMPORTANT?

- ✓ Reduces noise and inconsistencies.
- ✓ Improves model accuracy and performance.
- ✓ Helps machines focus on meaningful information.
- ✓ Essential first step before turning text into numbers.
- ✓ Better input → Better understanding → Better output.

Common Techniques

- ✓ Lowercasing
- ✓ Punctuation removal
- ✓ Tokenisation
- ✓ Stop word removal
- ✓ Stemming / Lemmatization
- ✓ Spelling correction
- ✓ Handling contractions and more...



KEY TAKEAWAY

Text processing turns messy, unstructured human language into **clean, structured data** that AI models can work with.



Next up:

Turning words into numbers (representations)!

STEP 2: UNDERSTANDING MEANING

Machines can't understand words directly. We convert them into numbers.



Computers work with numbers, not words.
So we represent words and sentences as **vectors (lists of numbers)** that capture their meaning.

“ Words that appear in similar contexts have similar meanings — and **similar** vectors! ”



FROM WORDS TO NUMBERS

Example sentence:

“I love natural language processing”



Tokens:

i love natural language processing



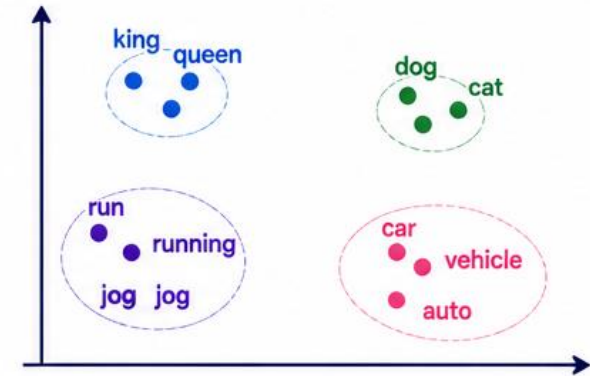
Each token is converted into a number vector (**embedding**)

WORD EMBEDDINGS (EXAMPLE)

Word	Vector Representation (Example)
cat	→ [0.23 0.81 0.12 ... -0.37 0.58]
dog	→ [0.25 0.79 0.10 ... -0.35 0.61]
king	→ [-0.31 0.42 0.75 ... 0.22 -0.66]
queen	→ [-0.29 0.40 0.73 ... 0.20 -0.63]
running	→ [0.11 0.32 0.21 ... -0.45 0.30]

 These numbers encode meaning, context and relationships between words.

SIMILAR MEANING = NEARBY VECTORS



Words with similar meanings appear closer together in the vector space.

TYPES OF WORD EMBEDDINGS



Count-based
(e.g., Bag of Words, TF-IDF)

Based on word frequency and co-occurrence.



Predictive
(e.g., Word2Vec, GloVe)

Learned from predicting or co-occurrence.



Contextual
(e.g., BERT, GPT embeddings)

Word representation depends on its context in a sentence.

KEY TAKEAWAY

- ✓ Words must be converted into numbers for machines to understand.
- ✓ Embeddings capture meaning and relationships between words.
- ✓ Word2Vec (and embeddings) will be covered later this semester in detail.



WHY IT MATTERS:

Embeddings are the foundation of modern NLP.
They allow models to understand meaning, not just words.



Next step: We will explore how embeddings are created (**Word2Vec**) in a later topic!

STEP 3: AI MODELS

NLP models learn patterns from data to understand, generate and use human language.



Over the years, NLP has **evolved** from simple rules to today's powerful AI models that understand **context** deeply.

“

The models may change, but the goal is the same: to understand **meaning** and **help** humans.

”



1. RULES & LINGUISTIC APPROACH (1950s – 1980s)



- Hand-crafted rules
- Grammar & dictionaries
- Expert knowledge

Example

If word ends with “ing”, it is a verb (heuristic rule)

Limitations

Hard to cover all rules.
Doesn't work for ambiguity or new situations.

2. MACHINE LEARNING APPROACH (1990s – 2000s)



- Models learn from data
- Statistical methods
- Feature engineering

Example

Spam classification using Naïve Bayes, SVM

Limitations

Needs manual features.
Struggles with large context and complex patterns.

3. NEURAL NETWORKS (2010s)



- Deep learning models
- Automatically learn features
- Handle more complex patterns

Example

RNNs, LSTMs for translation, sentiment, NER

Limitations

Sequential processing is slow.
Hard to capture long-range dependencies.

4. TRANSFORMERS (2017 – PRESENT)



- Self-attention mechanism
- Parallel processing
- Captures long-range context

Example

BERT, GPT, RoBERTa, T5, etc.

Limitations

Needs large data & compute.
Models can be resource intensive.

5. LARGE LANGUAGE MODELS (2020 – PRESENT)



- Trained on massive text data
- Understand and generate text
- Can follow instructions

Example

ChatGPT, Claude, Gemini, Llama, Kimi, etc.

Limitations

Hallucinations, bias, misuse risk, high cost, ethical concerns.

WHAT DO THESE MODELS DO?



Understand
e.g., sentiment, NER, intent



Translate
e.g., English to French



Summarise
e.g., long text to short text



Chat & Answer
e.g., chatbots, Q&A systems



Generate
e.g., stories, emails, code



Key Idea

We keep improving the models so they can better understand **humans** and help humans **better**.



KEY TAKEAWAY

From rules to large language models, NLP has evolved to understand **context**, handle complexity, and generate **human-like** responses.



Next up: Let's see how these models are used in real-world NLP applications!

FINAL OUTPUT: MAKING NLP USEFUL

After all the processing and modeling, NLP systems produce meaningful outputs.



Different tasks → different outputs

Same pipeline → **same goal**: convert human language into valuable results.

“

The pipeline may change for each task, but the purpose is always to produce **useful, intelligent and human-like output**.

”



1. INPUT

Human Language
(text or speech)



2. PROCESSING

Clean, tokenize,
represent, model



3. MODEL

AI model learns
patterns & context



4. PREDICTION

Model makes
predictions



5. OUTPUT

Useful result for
the user

EXAMPLES OF NLP OUTPUTS

Same pipeline, different tasks → different outputs



TRANSLATION

Convert text from one
language to another.

Input:

“Hello, how are you?”
(English)



Output:

“Bonjour, comment
ça va ?” (French)



SENTIMENT ANALYSIS

Detect the emotion or
attitude in text.

Input:

“The movie was
absolutely fantastic!”



Output:

Positive 😊
(Score: 0.92)



TEXT SUMMARISATION

Create a short summary of
a long text.

Input:

(Long article or report)



Output:

“A short, clear summary
capturing the main points.”



QUESTION ANSWERING

Answer questions based
on given information.

Input:

Q: What is NLP?
(Context provided)



Output:

“NLP is a field of AI that
helps computers understand,
interpret and generate
human language.”



CHATBOTS & CONVERSATION

Generate natural responses
in a conversation.

Input:

User: “What’s the
weather today?”



Output:

Bot: “It’s sunny ☀️
and 28°C today.”



TEXT GENERATION

Generate new text such as
stories, emails, code.

Input:

“Write a thank-you
email to a customer.”



Output:

“Dear Customer,
Thank you for your
purchase...”



KEY TAKEAWAY

NLP turns human language into intelligent outputs
that help us **communicate, decide, create and solve problems**.



Different tasks → different outputs
Same goal: making NLP useful for humans!

WHY NLP IS DIFFICULT

Human language is rich, messy and full of edge cases.



Natural Language is full of challenges that make it **extremely hard** for computers to understand, interpret and generate correctly.

“

Language is not black and white. It is full of exceptions!

”



CHALLENGE	WHAT IT MEANS	EXAMPLE
 Ambiguity	The same words or sentences can have multiple meanings.	“I saw her duck.” Did I see her lower her head, or her pet duck? 
 Context Dependence	Meaning depends on the surrounding words and situation.	“It’s cold.” Weather? Close the window? Unfriendly? 
 Sarcasm & Irony	The opposite of literal meaning is often intended.	“Great job!” Said with a smile vs. said with an eye-roll. 
 Idioms & Figurative Language	Figurative expressions don’t make literal sense.	“Break a leg!” Wishing good luck, not an injury! 
 Slang & Informal Text	Slang, abbreviations and new words change rapidly.	“That’s sick!” Cool? Impressive? Or actually unhealthy? 
 Spelling Variations & Errors	Humans make mistakes and write in many ways.	“definately”, “gonna”, “r u coming?” Different spellings, same intent. 
 Multiple Languages & Code-Mixing	Mixing languages and dialects is very common.	“Mo pe ale shop ek mo frer.” (Kreol + English) Which language rules apply here? 
 Emojis & Non-Verbal Cues	Emojis, tone, gestures add meaning beyond words.	😊😏😞 Same text, different emotions! 



KEY TAKEAWAY

NLP systems must handle ambiguity, context, variation and uncertainty at scale.

This is why building reliable NLP models is one of the **hardest problems in AI**.



Bottom line: Humans understand language effortlessly because we have brains, experience and common sense. Computers have to learn everything from data — and it’s never perfect.

THE LIMITATIONS OF NLP

*NLP has made great progress, but it is not perfect.
It still has important limitations we must understand.*



NLP systems can be powerful, but they do not think or understand like humans. Recognizing their limits helps us **use them wisely and responsibly**.

“

Knowing the limitations of NLP helps us set realistic expectations and build **better, more reliable** AI systems.

”



MAIN LIMITATIONS OF NLP



1 LACK OF TRUE UNDERSTANDING

NLP models predict the next word, but they do not truly understand meaning or intent like humans.

Example:
The model may give correct answers without really knowing the facts.

Impact:
Can lead to confident but incorrect responses.



2 DEPENDENT ON QUALITY DATA

NLP systems learn from data. If the data is biased, noisy or incomplete, the results will also be biased or wrong.

Example:
Biased training data can produce biased or unfair outputs.

Impact:
Reinforces bias and reduces trustworthiness.



3 STRUGGLE WITH CONTEXT

Understanding long texts, sarcasm, jokes, or cultural nuances is still very challenging.

Example:
May miss sarcasm and take a joke literally.

Impact:
Misinterpretation and inappropriate responses.



4 HALLUCINATIONS & INACCURACIES

Models can generate plausible but false or fabricated information with confidence.

Example:
Inventing facts, references, or statistics that do not exist.

Impact:
Spreads misinformation and reduces credibility.



5 PRIVACY & SECURITY RISKS

Sensitive data in text can be exposed or misused. Models may also be vulnerable to malicious inputs.

Example:
Leaking personal information or being tricked by prompt attacks.

Impact:
Threats to privacy, security and safety.



6 LIMITED GENERALISATION

Models may perform well in one domain but fail in unseen situations or for less-resourced languages.

Example:
Works well in English but struggles in low-resource languages.

Impact:
Inconsistent performance across tasks and domains.



WHAT THIS MEANS FOR REAL-WORLD APPLICATIONS



Human oversight is essential.



Ethical design and fairness must be prioritized.



Fact-checking and verification are always necessary.



NLP should support humans, not replace human judgment.



Responsible use builds trust and encourages safe adoption.



KEY TAKEAWAY

NLP is an amazing technology, but it is not perfect. Understanding its limitations helps us build systems that are **reliable, ethical and human-centered**.



REMEMBER

- NLP models generate predictions, not absolute truth.
- The better we understand the limits, the better we can design, evaluate and improve NLP systems.



THE CHALLENGES IN NLP

NLP has come a long way, but it still faces many real-world challenges.



Human language is **complex**, **dynamic** and **context-dependent**.
These challenges make it hard for machines to fully understand and respond like humans.

“

The more we understand the challenges, the better we can build NLP systems that are **useful**, **fair** and **responsible**.

”



KEY CHALLENGES



1 AMBIGUITY

Words and sentences can have multiple meanings depending on context.

Example: “Bank” can mean a financial institution or river bank.



2 CONTEXT UNDERSTANDING

Understanding the full context of a sentence, conversation or situation is still difficult.

Example: “I saw her duck” (a bird or a movement?).



3 LANGUAGE DIVERSITY

NLP models struggle with low-resource languages, dialects and code-switching.

Example: Creole, regional languages, mixed languages.



4 DATA QUALITY & BIAS

Biased or unbalanced data leads to unfair or inaccurate model predictions.

Example: Biased sentiment analysis on certain groups.



5 PRIVACY & SECURITY

Handling sensitive text data raises concerns about privacy, security and misuse.

Example: Chat logs, medical records, personal data.



6 HALLUCINATIONS

Models can generate plausible but incorrect or made-up information.

Example: Providing fake facts confidently in answers.



7 LIMITED REASONING

Models find it hard to reason, plan or solve complex problems step by step.

Example: Math word problems, multi-step logic tasks.



8 COMPUTATIONAL COST

Training large models requires huge computing power, energy and financial resources.

Example: Training large LLMs can cost millions of dollars.



9 FAIRNESS & ETHICS

Ensuring fairness, inclusivity and alignment with human values remains challenging.

Example: Gender bias, stereotypes or harmful content.



10 ADAPTABILITY OVER TIME

Language evolves quickly; models can become outdated or irrelevant.

Example: Slang, new words, trends, world events.



KEY TAKEAWAY

These challenges remind us that NLP is not just a technical problem—it is also a **human**, **social** and **ethical** challenge.
We must design NLP systems that are **accurate**, **fair**, **safe** and **trustworthy**.



FUTURE DIRECTION

- Develop more inclusive and diverse data
- Improve reasoning and factual grounding
- Build responsible, transparent and secure AI
- Keep humans at the centre of NLP systems



Understanding the challenges helps us build better NLP systems—for everyone.

Next: Part 7 – Future of NLP →

THE FUTURE OF NLP

NLP is evolving rapidly and will continue to transform how humans and machines communicate.



The next decade of NLP will bring more **intelligent**, **multimodal** and **autonomous** systems that understand us better and help us do more.

1 AI AGENTS & AUTONOMOUS ASSISTANTS

NLP will power agents that can plan, reason and take actions to complete tasks for us.

- Personal assistants
- Task automation
- Smart workflows



2 MULTILINGUAL & CROSS-CULTURAL AI

Models will understand and generate more languages and dialects with cultural awareness and sensitivity.

- Better translation
- Low-resource languages
- Inclusive communication



3 SPEECH-TO-SPEECH AI

Seamless spoken conversations with real-time understanding and natural voice responses.

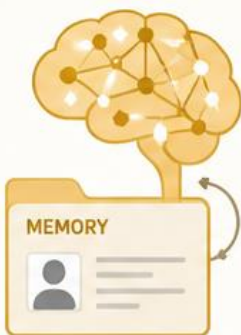
- Voice assistants
- Real-time interpretation
- Voice cloning & synthesis



4 LONG-TERM MEMORY & PERSONALIZATION

Models will remember more complex information and personal preferences over time.

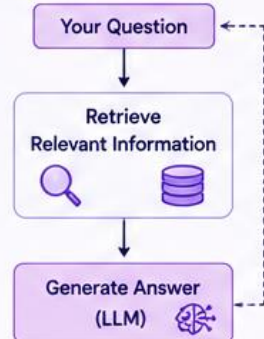
- Persistent memory
- Personalized experiences
- Better continuity



5 RETRIEVAL-AUGMENTED GENERATION (RAG)

Combining knowledge retrieval with generation for more accurate, up-to-date and trustworthy responses.

- Connect models to knowledge
- Reduce hallucinations
- Dynamic, factual answers



6 TOWARDS GENERAL INTELLIGENCE

Moving towards systems that can understand, reason and learn across many domains like humans.

- Better reasoning
- Transfer learning
- Lifelong learning



NLP is not just about understanding language.

It is about building intelligent systems that communicate, collaborate and create value for humanity.

THE JOURNEY CONTINUES...

As data grows, models improve, and possibilities expand.



BUILDING THE FUTURE: OUR CALL TO ACTION

The future of NLP depends on how we build, use and govern these powerful technologies.



We all have a role to play—researchers, developers, organizations, policymakers and users—in shaping **NLP for the greater good**.

WHAT WE MUST DO



1. HUMAN-CENTERED DESIGN

- Put people at the center.
- Design NLP systems that are useful, inclusive and respect human dignity.
- Consider diverse needs and contexts.



2. RESPONSIBLE DEVELOPMENT

- Build models that are safe, reliable and robust.
- Test for fairness, bias and real-world impact.
- Be transparent about capabilities and limits.



3. TRANSPARENCY & ACCOUNTABILITY

- Make decision-making more explainable.
- Establish clear accountability for AI outcomes.
- Enable auditability and user recourse.



4. PROTECT PRIVACY & SECURITY

- Protect user data and conversations.
- Follow strong privacy by design principles.
- Secure models against abuse and misuse.



5. COLLABORATE FOR IMPACT

- Foster interdisciplinary collaboration.
- Share knowledge responsibly.
- Work together to solve real human problems.



KEY TAKEAWAY

Great NLP systems are not just about technology—they are about **trust, ethics and positive impact** for individuals, communities and society.



OUR VISION

NLP technologies that empower people, respect values and improve lives everywhere.



OUR RESPONSIBILITY

To build, use and govern NLP with integrity, empathy and long-term thinking.



OUR FUTURE

A world where humans and NLP work together to create a better, more inclusive and intelligent future.



Let's build the future of NLP—together.
Responsible today, better tomorrow.



The choices we make today will shape the NLP systems of tomorrow.



Let's create NLP systems that are **intelligent, trustworthy** and truly **human-centric**.

